Adith Augustine TJ

Data Engineer

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SUMMARY

Results-driven Data Engineer with 4.7 years of experience in designing **scalable data pipelines**, optimizing **ETL workflows**, and working with **cloud-based solutions**. Skilled in **Python, SQL, PySpark, and Azure Databricks**, with a proven track record of **enhancing data processing efficiency and automation**. Successfully **migrated data for 7+ ERP systems**, ensuring **100% transformation accuracy** and improving **processing time by 70%** through performance optimizations. Passionate about **building reliable, high-performance data solutions** that drive business insights and decision-making.

PROFESSIONAL EXPERIENCE

Data Engineer Nov '20 - Present

Tata Consultancy Services Limited Bangalore, India

Big Data Processing and Analytics (*Life Sciences and Healthcare)*

- **Developed and optimized automated test scripts** using **PySpark** and **Spark SQL** to validate data integrity in **Azure Databricks pipelines**, reducing manual validation efforts by **80%.**
- **Created and implemented scalable ETL pipelines** and data transformation workflows using *PySpark, SQL, and Databricks* to process **high-volume structured and unstructured data**, ensuring **100% data transformation accuracy** through robust testing and validation.
- Built and **scheduled Databricks workflows** for automated data ingestion and transformation, ensuring timely and reliable data availability across environments.
- Designed **optimization scripts** to efficiently process **high-change CDC tables**, improving data refresh performance and reducing latency.
- Successfully **migrated data for 7+ ERP systems**, including **full and incremental (CDC) loads**, ensuring compliance with **SOX** and **non-SOX** requirements. Automated configurations for different ERPs, enabling seamless data ingestion from on-prem to cloud storage layers while ensuring **100%** data accuracy.
- Utilized **Azure Data Lake Storage Gen2 (ADLS)** and Blob Storage for scalable data storage, integrating with **Snowflake** for **high-performance querying** of sensitive data.
- Developed automated workflows for **real-time and historical data ingestion**, implementing scheduled updates every 15 minutes for time-sensitive datasets.
- Created Python- and Spark-based **validation and reconciliation scripts** to identify and resolve data discrepancies between **bronze and silver layers**, improving data consistency.
- Implemented Databricks optimization techniques, including **Delta Lake optimizations** , resulting in a **70%** reduction in processing time while reducing storage costs and enhancing query performance.
- Contributed to the development and optimization of data processing workflows for data warehousing projects and leveraged PySpark and SQL expertise to **increase data processing speeds by 30%.**
- Implemented **data governance** practices by writing ingested data from the **silver layer** to **Unity Catalog**, ensuring **secure**, **discoverable**, and **well-governed** data access..
- Implemented **parameterization** in automated scripts to enhance **flexibility** and **reusability,** allowing **dynamic configuration** of pipeline activities based on runtime parameters and input values.

Translational Services

- Developed a Python script to retrieve data from **Amazon Redshift** and perform automated language translation using **AWS APIs** and **dictionary-based lookups.**
- Designed a **configuration-driven approach** to update translation dictionaries and integrate with **user-defined schemas dynamically**.
- Optimized performance using **multiprocessing** and **multithreading**, reducing translation processing time by over **85%**.

Hyper API

- Designed a Python-based POC to **optimize Tableau data processing** by offloading CSV extraction from S3 to an Azure VM, reducing execution time from **5+ hours to under 2 hours**.
- Automated the creation of Hyper files and publishing them to Tableau Server, significantly **cutting operational costs**.
- Developed error notification mechanisms, file encryption, and command-line **argument parsing** for enhanced automation.

Project Management and Agile Methodology

- Embraced **Agile methodology**, employing **Jira** for seamless planning and execution of project activities. This approach fostered collaboration, adaptability, and efficient project management, resulting in successful delivery and **customer satisfaction**.
- Led **Bitbucket** code development, fostering **team collaboration**, enhancing productivity, and ensuring efficient project management with robust version control.

Data Science Intern Jan '20 - Apr '20

Decathlon Data Team (India) Bangalore,India

- Created a **product recommendation system** using **image similarity techniques**, enhancing personalized customer experiences.
- Created **interactive logistics visualizations** using **Folium** with data extracted from **QlikSense**, optimizing supply chain insights.

SKILLS

Cloud Platforms- Microsoft Azure, Azure Databricks, AWS
Programming Languages: Python, Pyspark, SQL
Data Warehousing- SQL, Azure SQL Databases, Snowflake
Tools & Technologies- ADF, Azure Datalake Storage, Azure Blob Storage,Azure Storage Account
Agile: Jira, Confluence
Version Control: Git

EDUCATION

Masters of Science in Big Data Analytics Jun '18 - Apr '20

St. Joseph’s College Autonomous Bangalore, India

CGPA: 7.95/10

Bachelor of Science in Mathematics, Physics,Statistics Jun '15 - Apr '18

Sacred Heart College Autonomous Kochi, Kerala

CGPA: 8.13/10

CERTIFICATIONS

- DP-203 Microsoft Azure Data Engineer
- AZ-900: Microsoft Azure Fundamentals
- AI-900: Microsoft Azure AI Fundamentals
- AI Engineer (TCS Internal Program, 1 year)

AWARDS & RECOGNITION

- On the Spot(Team) Award - MAR'25, DEC'24, NOV'22
- Best Team Award - FEB'25, JUL'24
- Service & Commitment Award- NOV'23
- Xcelerate Warrior Certificate (Appreciation)- Dec'22
- Fresco Play Miles Award- DEC'21
- Innovation in Science Pursuit for Inspired Research (**INSPIRE**) Scholarship : Department of Science and Technology, Central Goverment

Hobbies and Interests

Dance (TCS Dance team), Football , Gym , Swimming, Badminton, Aptitude Preparation